

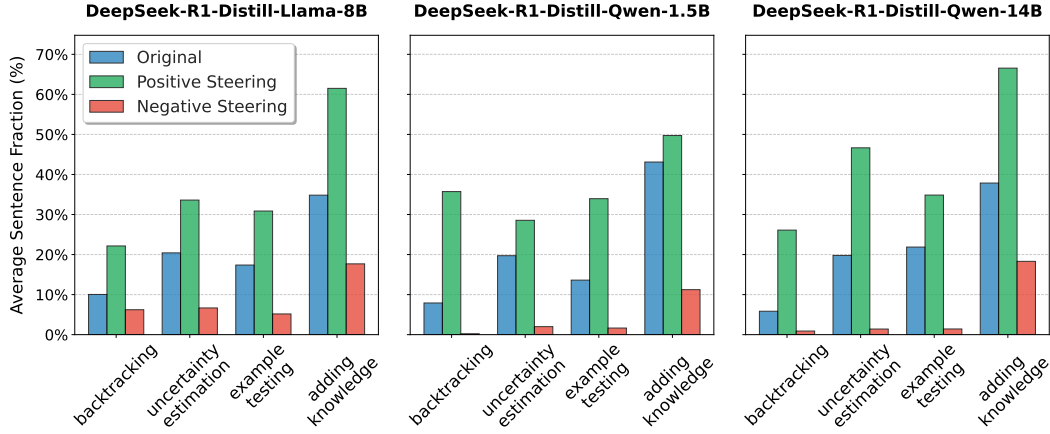


Figure 4: Effect of applying the steering vector for each reasoning behavior across different distill models. The y-axis shows the change in the fraction of tokens exhibiting each behavior when applying positive or negative steering. Positive steering increases behaviors such as backtracking and uncertainty estimation, while negative steering suppresses or significantly reduces them, confirming the causal influence of our extracted vectors.

4.3 SELECTING FINAL STEERING VECTORS

To determine the final steering vectors, we apply *attribution patching* (Section 2.1) to quantify the causal relevance of each vector in its respective layer. Specifically, we consider the following patching experiment: Given a candidate steering vector $\mathbf{u}_\ell^c$ for a specific behavioral category, we add it to the residual stream activation preceding a token-sequence annotated with one of the relevant behaviors. Therefore, we define the patched activation as:

$$\mathbf{a}_\ell^{\text{patched}} = \mathbf{a}_\ell + \mathbf{u}_\ell^c.$$

If this intervention leads to a significant change in the KL divergence of the next-token prediction, then the steering vector in layer ℓ is *causally relevant* for the given behavior. We approximate the patching effect for this experiment with:

$$\Delta L \approx (\mathbf{u}_\ell^c)^T \cdot \frac{\partial}{\partial \mathbf{a}_\ell} L(\mathbf{x}_{\text{clean}} \mid \text{do}(\mathbf{a}_\ell = \mathbf{a}_{\text{clean}})),$$

where $\mathbf{u}_\ell^c = (\mathbf{a}_\ell^{\text{patched}} - \mathbf{a}_\ell)$. We average the absolute patching effect for each category over all category-sequences in all 500 reasoning chains. The results are shown in Figure 3. Based on these results, we can select the causally most relevant steering vectors from the layers where the patching scores are highest. However, we avoid selecting early layers that show excessive correlation with embedding tokens, as these layers primarily capture token-specific representations rather than behavioral patterns (Appendix D).

4.4 EVALUATION OF STEERING VECTORS

To evaluate the effectiveness of our extracted steering vectors, we apply them at the selected layers (see Table 2) and observe their influence on the model’s reasoning process. Steering is implemented by *adding or subtracting* the extracted steering vectors $\mathbf{u}_\ell^c$ to the residual stream activations at inference time. By applying this intervention, we can increase or decrease behaviors such as backtracking, uncertainty estimation, and example testing, providing a direct mechanism for manipulating the model’s reasoning process.

4.5 RESULTS OF STEERING

We apply each steering vector to 50 unseen reasoning tasks and analyze how the model’s reasoning behavior changes. The results, presented in Figure 4, demonstrate that our extracted vectors effectively

Behavioral Category	DeepSeek-R1-Distill Llama-8B	DeepSeek-R1-Distill Qwen-1.5B	DeepSeek-R1-Distill Qwen-14B
Uncertainty Estimation	12	18	29
Example Testing	12	15	29
Backtracking	12	17	29
Adding Knowledge	12	18	24

Table 2: Selected layers for each behavioral category based on attribution patching results. For each model, we select the layer with the maximum score from the attribution patching experiments, ignoring early layers that are highly correlated with embedding tokens.

control the model’s reasoning patterns. Appendix E includes a full example of positive and negative steering on the “Adding Knowledge” vector for the DeepSeek-R1-Distill-Qwen-14B model.

As shown in Figure 4, positive steering increases behaviors such as backtracking, uncertainty estimation, and example testing, while negative steering reduces them. These effects are consistent across both DeepSeek-R1-Distill models, reinforcing the hypothesis that Thinking LLMs encode these reasoning mechanisms as linear directions in their activation space. Our findings confirm that steering vectors provide a reliable and efficient method for interpreting and controlling the internal reasoning dynamics of thinking large language models. Further analysis of the cosine similarity between different steering vectors reveals that most behavioral categories correspond to distinct directions in the model’s activation space (see Appendix C).

5 RELATED WORK

Recent work has explored methods for steering and interpreting language models by identifying meaningful directions or features within their internal representation spaces. Subramani et al. (2022) show that extracting latent steering vectors from pretrained language models can systematically alter the generated text. Similarly, Turner et al. (2023) propose *activation engineering*, modifying model activations at inference time to control outputs, in contrast to prompt engineering or fine-tuning. Extending this line of research, Panickssery et al. (2023) introduce Contrastive Activation Addition (CAA), which derives a “contrastive” vector by averaging differences in residual stream activations between positive and negative examples of a target behavior. Adding this vector to a model’s activations elicits more desirable outputs without retraining. Beyond these methods, Zou et al. (2023) propose *Representation Engineering*, offering a top-down approach to refining and analyzing internal representations for greater transparency.

Applying these methods for fine-grained control of language models through systematic manipulation of their internal representations Li et al. (2023) propose an *inference-time intervention* to encourage more truthful responses without additional training. Additionally, Ardit et al. (2024a) has shown that refusal behavior can be localized to a single direction in latent space, that minimally affect other capabilities, enabling targeted interventions to encourage refusal or jailbreak the model.

A related line of work focuses on leveraging these internal representations for reasoning and chain-of-thought. Zhao et al. (2025) found a steering vector to efficiently elicit long chain-of-thought in language models. Dutta et al. (2024) analyzed the neural sub-structures within Llama-2 7B that facilitate multistep reasoning over fictional ontologies and found that the model deploys multiple parallel pathways for step-by-step reasoning. In concurrent work, Hazra et al. (2025) train sparse autoencoders on DeepSeek’s 671B-parameter R1 reasoning model, uncovering internal features associated with reasoning behaviors, such as backtracking, which can be used for steering interventions.

6 CONCLUSION AND FUTURE WORK

This work presents a steering approach for controlling reasoning behaviors in thinking LLMs, with a specific focus on DeepSeek-R1-Distill models. We demonstrate that several reasoning behaviors exhibited by these models, including expressing uncertainty, backtracking, and example testing, can be effectively controlled using steering vectors. While our analysis does not claim to provide

a complete taxonomy of reasoning mechanisms, it establishes a practical framework for steering specific aspects of model behavior. Our key findings indicate that:

- Several reasoning behaviors in thinking models can be isolated to specific directions in the model’s activation space, enabling precise control through steering vectors.
- Our steering approach is effective across a diverse set of 500 tasks, demonstrating robust control over targeted reasoning behaviors.
- The method generalizes across different model architectures within the DeepSeek-R1-Distill family, showing consistent steering effects.

These results provide practical tools for modulating reasoning capabilities in thinking models. The ability to adjust specific aspects of the reasoning process through steering vectors opens new possibilities for adapting these models to different tasks and requirements.

Despite these promising results, our work has several limitations that suggest directions for future research. The current automated annotation process using GPT-4o, while efficient, has occasionally produced false positives and false negatives in identifying reasoning patterns. Future work should focus on developing more robust annotation methods, potentially incorporating multiple models or human validation to improve accuracy. Additionally, while our analysis centers on DeepSeek-R1-Distill models, the generalization of these findings to other models that have undergone reward-based RL training instead of being fine-tuned on thinking models (e.g., Qwen’s QwQ) remains an open question. Extending this research to a broader range of models would provide deeper insights into the universality of thinking mechanisms and their practical applications. By addressing these limitations, future research can further advance the understanding and control of thinking models, paving the way for more reliable and adaptable AI systems.

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REFERENCES

- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in Language Models Is Mediated by a Single Direction. *arXiv*, June 2024a. doi: 10.48550/arXiv.2406.11717.
- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv*, abs/2406.11717, 2024b. URL <https://arxiv.org/abs/2406.11717>.
- François Chollet. OpenAI o3 Breakthrough High Score on ARC-AGI-Pub, December 2024. URL <https://arcprize.org/blog/oai-o3-pub-breakthrough>.
- DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement ‘learning, 2025.
- Subhabrata Dutta, Joykirat Singh, Soumen Chakrabarti, and Tanmoy Chakraborty. How to think step-by-step: A mechanistic understanding of chain-of-thought reasoning, 2024. URL <https://arxiv.org/abs/2402.18312>.
- GDM. Gemini flash thinking: Gemini 2.0 Flash Thinking Experimental, 2024. URL <https://deepmind.google/technologies/gemini/flash-thinking/>.
- Dron Hazra, Max Loeffler, Murat Cubuktepe, Levon Avagyan, Liv Gorton, Mark Bissell, Owen Lewis, Thomas McGrath, and Daniel Balsam. Under the hood of a reasoning model, 2025. URL <https://www.goodfire.ai/blog/under-the-hood-of-a-reasoning-model>. Accessed: 2025-06-22.